Answer ALL Questions

1. Question on Wind and Photovoltaic Power

a) The wind speeds before and after the plane of rotation of an ideal wind turbine are v_1 and v_2, respectively.

i) Show that the maximum power is captured by the wind turbine when v_2 is one-third of v_1. Make the simplifying assumptions as necessary.

[4]

Assuming a constant mass flow rate of air

$$\dot{m} = \rho A_1 v_1 = \rho A v' = \rho A_2 v_2$$

$$P_w = \frac{1}{2}\rho A_1 v_1^3 - \frac{1}{2}\rho A_2 v_2^3 = \frac{1}{2}\dot{m}(v_1^2 - v_2^2)$$

[1 mark]

$$F = \dot{m}(v_1 - v_2)$$

$$P_w = Fv' = \dot{m}(v_1 - v_2)v'$$

[1 mark]

$$\Rightarrow v' = \frac{v_1 + v_2}{2}$$

$$P_w = \frac{1}{2}\rho A(v_1^2 - v_2^2)\left(\frac{v_1 + v_2}{2}\right)$$

[1 mark]

$$C_P = \frac{P_w}{P_A} = \frac{1}{2}\left[1 - \left(\frac{v_2}{v_1}\right)^2\right]\left[1 + \frac{v_2}{v_1}\right] = \frac{1}{2}(1 - k^2)(1 + k)$$

$$\frac{dC_P}{dk} = 0 \rightarrow k = \left(\frac{v_2}{v_1}\right)_{\text{opt}} = \frac{1}{3}$$

[1 mark]

(bookwork – use of standard derivation)

Some students assumed the final form of the expression for P_w without deriving it.

ii) Use the above to calculate the theoretical maximum possible efficiency of power extraction from the wind.

[1]

$$C_{P-\max} = \frac{1}{2}\left[1 - \left(\frac{v_2}{v_1}\right)_{\text{opt}}^2\right]\left[1 + \left(\frac{v_2}{v_1}\right)_{\text{opt}}\right] = 0.593$$

(bookwork – continuation of previous derivation)

Model Answers

b) Explain why variable speed operation is desirable for wind turbine generators.

[3]

- The efficiency of a wind turbine depends on the tip-speed ratio $\lambda = \frac{\omega_T R}{v_w}$ and is maximum at a given $\lambda = \lambda_{\text{opt}}$
- For different wind speeds, the rotational speed of the wind turbine has to vary according to $\omega_T = \frac{\lambda_{opt} \times v_w}{R}$ to keep operating at the optimum tip-speed ratio and thus, maximise the power output.
- Apart from maximising power capture, variable speed operation reduces the mechanical stress on the wind turbines wind speed varies.

[1 mark for each point]

Several students misinterpreted the question and described different regions of the turbine (power vs. wind speed) characteristics. Some strayed into wind speed distribution and energy capture without referring to the fact that the efficiency is maximum for an optimum TSR. Some discussed the forces acting on a wind turbine in detail which was not necessary. Only a few students covered the last point on reduced mechanical stress.

(bookwork – interpretation of material from lecture notes)

c) A wind turbine with 80 m long blades has a maximum efficiency of 48% at the optimum tip-speed ratio of 7.5. Neglecting the losses in the generator and gearbox and assuming an air density of 1.22 kg/m^3, calculate the following:

i) the maximum power that can be generated (in MW) at a wind speed of 8 m/s.

[1]

$A = \pi R^2 = 20{,}106 \text{ m}^2$

$P_w = \frac{1}{2} \rho A v_w^3 C_{p-opt} = 3.014 \text{ MW}$

ii) approximate mass flow rate (kg/s) of air through the wind turbine under the above condition in (i).

[2]

$\rho A = 24{,}530 \text{ kg/m}$

$\dot{m} = \rho A \left(\frac{v_w + v_2}{2}\right) = \frac{2}{3} \rho A v_w = 1.31 \times 10^5 \text{ kg/s}$

Some students used the incoming wind speed v_w to calculate mass flow rate

Model Answers

iii) the rotational speed of the turbine (in rpm) to capture the maximum power at a wind speed of 10 m/s.

[2]

$$\lambda = \frac{\omega_T R}{v_w} \rightarrow \omega_T = \frac{\lambda_{opt} \times v_w}{R} = 0.938 \text{ rad/s} = 8.95 \text{ rpm}$$

d) A group of 24 solar PV modules are to be installed at a location with a latitude 35°N. Each module has a rated power of 300 W with 15% efficiency under standard test condition (STC). The terminal voltage of the PV module is 55 V for 300 W output under STC. A PV array is formed with 4 parallel strings with each string having 6 such PV modules in series with a string diode.

i) What is the approximate surface area (in m^2) of each PV module?

[1]

$$A = \frac{P_{\text{rated}}}{1000 \times \eta} = 2 \text{ m}^2$$

Some students an incorrect value of solar irradiance at STC (e.g. 800 W/m^2, 1100 W/m^2)

ii) What should be the tilt angle (in degrees with respect to the horizontal) of the PV modules at solar noon if the declination angle on a particular day is 20°?

[2]

At solar noon
$\alpha = 90° - (\phi - \delta)$

Irradiance on PV module
$I_{\text{module}} = I_{\text{incident}} \sin(\alpha + \beta)$

To maximise irradiance on PV module
$\alpha + \beta = 90° \rightarrow \beta = \phi - \delta = 15°$

iii) What is the maximum total power output of the PV array when the irradiance is 500 ~~k~~W/m^2 assuming the same efficiency as under STC?

[2]

$$P_{\text{arr}} = N_{\text{mod}} \times 500 \times A \times \eta = 3.6 \text{ MW}$$

Typo – should have been 500 W/m^2 in which case the answer should be 3.6 kW, but students got full marks for obtaining either 3.6 kW or 3.6 MW although the latter is not practical

iv) How much power can be generated by the PV array under STC if two PV modules in separate strings are completely shaded?

Model Answers

[1]

The current fed by both the affected strings would be zero as the corresponding string diodes are reverse biased. Hence power output under STC will be

$$P_{\text{out}} = N_{\text{mod}} \times 300 \times \frac{2}{4} = 3.6 \text{ kW}$$

v) How much power can be generated by the PV array under STC if two PV modules within the same string are completely shaded?

[1]

The current fed by one affected string with two shaded modules would be zero. Hence power output under STC will be

$$P_{\text{out}} = N_{\text{mod}} \times 300 \times \frac{3}{4} = 5.4 \text{ kW}$$

Mistake in (i) trickled into (iii) – (v) for some students

(problem solving – use of standard formulae, deduction and interpretation)

Model Answers

2. Question on balancing, Voltage Control and Security of Supply

Part A

a) Explain why large-scale penetration of wind generation will impose requirements for flexibility and explain why the need for "reserve" will increase.

[1]

Management of uncertainties in wind production require increase in reserve to be able to balance demand and supply

b) Explain the advantages and disadvantages of providing reserve through "spinning" and "standing" reserve technologies.

[3]

- Reserve provided by synchronised (fossil fuel) plant (running part loaded)
  - Spinning reserve – part loaded conventional generation can deliver reserve in short time scale
  - There is reduction in efficiency of part loaded plant (20%CCGT and 10-15% coal) and increased emissions - energy delivery accompanies provision of reserve, which may lead to curtailment of renewables

- Reserve provided by standing plant
  - Standing reserve will be used only when needed
  - Generating plant that provides standing reserve (OCGT) generally is characterised by higher fuel cost and the it will need time to provide reserve; On the other hand, DSR and Storage can provide standing reserve with no impact

Part B

c) Explain how on-load tap-changer voltage control could contribute to the amount of distributed generation that could be accommodated in a distribution network.

[3]

The answer along the following lines will give full marks. Coordinated voltage control using on load tap changer could potentially, in general, **increase** the amount of distributed generation that could be accommodated in a distribution network by:

- Decreasing voltage during high generation and low demand conditions, and
- Increase voltage during low generation and high demand conditions.

d) What will be the impact of active voltage control on network losses?

[1]

Enhancing the utilisation of network assets by on-load tap-changer voltage control will increase network losses.

Model Answers

Part C

Contribution of distributed generation to the security of supply depends on the demand profiles and availability of the resources. In this case, the plant contains 3 units that can provide network support. Two of those are the same, each having rating of 10 MW and technical availability of each unit is 80%. Rating of the third unit is 20 MW with technical availability of 90%.

Demand profile during a peak season is represented by two characteristic days. Normalised load duration curves of these two characteristic days are shown in Figure 2.2.

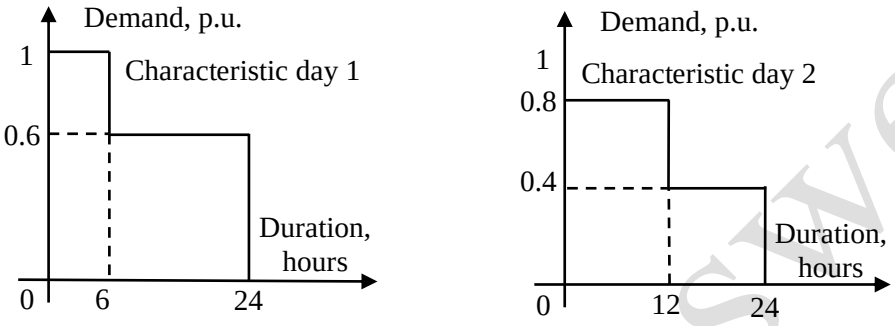


Figure 2.2 Normalised load duration curves for two characteristic days

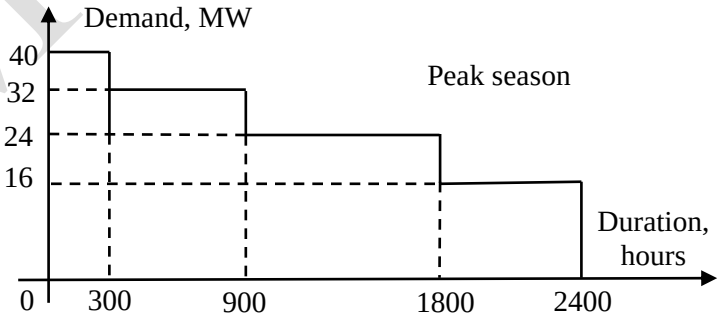
During a peak season there are:

- 50 days for which load curve can be represented by using Characteristic day 1
- 50 days for which load curve can be represented by using Characteristic day 2

e) Sketch the peak season load duration curve that includes both peak day type, that will be used to calculate the security contribution of the distributed generation plant. When calculating the contribution to the security of supply, scale demand to the total generation plant capacity.

[2]

The peak season load duration curve diagram, used to calculate contribution of the generation plant, is



The peak demand is equal to the total generation plant capacity of 40 MW and the peak season duration is 2400 hours.

Model Answers

f) Form the capacity outage probability table for this plant, and calculate the corresponding probability value

[4]

Capacity outage probability table for the generation plant is

Capacity in service, MW	Probability
40	$0.8^2 \times 0.9 = 0.576$
30	$2 \times 0.8 \times 0.9 \times (1-0.8) = 0.288$
20	$0.8^2 \times (1-0.9) + (1-0.8)^2 \times 0.9 = 0.1$
10	$2 \times 0.8 \times (1-0.8) \times (1-0.9) = 0.032$
0	$(1-0.8)^2 \times (1-0.9) = 0.004$

The total probability for all five possible capacity states is 1.

[4 marks; subtract 1 mark for any missing capacity in service and wrong probability up to a maximum of subtracted 4 marks]

g) Calculate the Expected Energy Not Supplied of the system supplied by the generation plant using above load duration curve.

[3]

Expected energy not supplied is

Capacity, MW	Probability	EENS
40	0.576	0
30	0.288	$(10 \times 300 + 2 \times 600) \times 0.288 = 1209.6$
20	0.1	$(20 \times 300 + 12 \times 600 + 4 \times 900) \times 0.1 = 1680$
10	0.032	$(30 \times 300 + 22 \times 600 + 14 \times 900 + 6 \times 600) \times 0.032 = 1228.8$
0	0.004	$(40 \times 300 + 32 \times 600 + 24 \times 900 + 16 \times 600) \times 0.004 = 249.6$

Total EENS is 4368 MWh.

h) Calculate the capacity of the equivalent source, firm capacity, that never fails that results in the same Expected Energy Not Supplied.

[3]

EENS for firm capacity of 32 MW is
$EENS_{32MW} = 8 \times 300 = 2400$ MWh

EENS for firm capacity of 24 MW is
$EENS_{24MW} = 2400 + 8 \times 900 = 9600$ MWh

Firm capacity that results in the same EENS, generation plant contribution in MW, is between 24 and 32 MW.
EENS = $2400 + DC \times 900 = 4368$ MWh
DC = 2.187 MW
C = 32 – DC = 29.813 MW
Contribution in MW is 29.813 MW.

Model Answers

3. Question on Low Carbon Heat

a) Give a brief description of the following types of low carbon heating systems.

i) Air source heat pump

[2]

ASHP transfers thermal energy from the outside air to the area to be heated. There are 4 stages to the cycle and these comprise:

- Evaporation - The refrigerant passes through a heat exchanger where it evaporates absorbing heat in the process.
- Compression - The heated refrigerant is then compressed to higher pressure using an electrically driven compressor.
- Condensation - The compressed refrigerant passes through another heat exchanger located in the space to be heated and condenses, changing state back to a liquid and rejecting heat in the process.
- Expansion – The liquid refrigerant passes through an expansion device lowering its pressure prior to returning to the evaporator.

ii) Hybrid heat pump

[2]

This operates on the same principle as an ASHP except it incorporates an additional heating source such as a condensing gas boiler to provide further heat capacity for both space and water heating

iii) District heating

[2.5]

District heating is a system where heat for more than one building or an area is produced at a central location and distributed through a network of insulated pipes. Heat sources include:

- Combined heat and power units
- Network heat pumps
- Waste(d) heat recovery
- Solar thermal
- Geothermal

The main components of a district heating system are:

- Heat storage - to provide additional capacity and backup, etc.
- Heat network with pumps – connecting heat sources with buildings using supply and return pipes.

iv) Hydrogen heating

[2.5]

This is based on the repurposing of the existing gas distribution network from natural gas to hydrogen. Hydrogen can be produced using a number of processes but at present the

Model Answers

most suited for large scale production is steam methane reforming using natural gas with carbon capture and storage and electrolysis using low carbon electricity.

b) In 2019 the UK introduced legislation to bring all greenhouse gas emissions to net-zero by 2050. This compares to its previous targets of at least 80% reduction from 1990 levels. In order to meet this target, it is likely that the removal of CO2 from the atmosphere will be required. Briefly explain why?

[2]

This is because there are sectors of the economy described as "hard-to-treat" where at present there are limited options available to reduce GHG emissions. For example, aviation and agriculture.

c) Two technologies for removing CO2 from the atmosphere are:

i) Bioenergy with carbon capture and storage (BECCS)

ii) Direct air capture of CO2 with storage (DACCS)

Give a brief description of each

[4]

BECCS
This converts biomass into energy. It is regarded as having "net neutral" CO2 emissions as the biomass is only releasing the CO2 absorbed when it was a crop. The CO2 emissions are captured using a chemical process and then transported to permanent storage in a geological reservoir. Hence overall the emissions are regarded as negative.

DACCS
Direct air capture of CO2 uses chemical processes to remove CO2 from the atmosphere. The CO2 can then be stored permanently in a geological reservoir. Its main drawback is that at present it is very energy intensive.

d) A householder has decided to install a low carbon heating system and is trying to decide whether to convert their gas boiler to a hydrogen boiler or to purchase an air source heat pump (ASHP) system. The data are as follows:

	Heating system fixed costs	Running costs	Efficiency
Hydrogen boiler	£250pa	£0.18/kWh	90%
Air source heat pump	£1000pa	£0.15p/kWh	300%

(The fixed costs include the cost of the installation allocated over the life of the heating system.)

i) What is the annual heat demand whereby the total annual costs of the ASHP is the same as the H2 boiler?

[4]

Model Answers

Need to find the heat demand, X, when:

Cost of hydrogen heating = cost of ASHP heating

250 + (0.18/90%) *X =1000 + (0.15/300%) *X
X*(0.2-0.05) = 750
X=750/0.15
X=5,000 kWh pa

Hence, the total annual costs are the same when the heat demand is 5,000kWh pa

ii) The householders' demand is 6,000kWhpa. Which system has the lowest annual cost?

[1]

The ASHP.

Model Answers

4. Question on Flexibility, Hydro Power and Offshore Transmission

Demand Flexibility

A low-voltage feeder that is supplying a residential area, has a baseline (non-EV) demand profile peaking at 250 kW on a winter day, as shown in Figure 4.1. Due to favourable government incentives, 30 EVs have been purchased by the residential customers in the area. As specified in Table 4.1, they can be divided into 3 groups according to their driving patterns i.e. the time they arrive home and State of Charge (SoC) at arrival. Each EV has a 30 kWh battery, while the rating of each EV charger is 6 kW (for both charging and discharging, if allowed). Demand can be assumed constant in each 1-hour period. Ignore any charging or discharging losses. Answer the following questions:

a) If all EV users plug in their vehicles to charge immediately upon arrival and charge them at maximum capacity until their batteries are full, what is the total peak demand on the feeder and when does it occur? Plot the new total demand diagram.

[4]

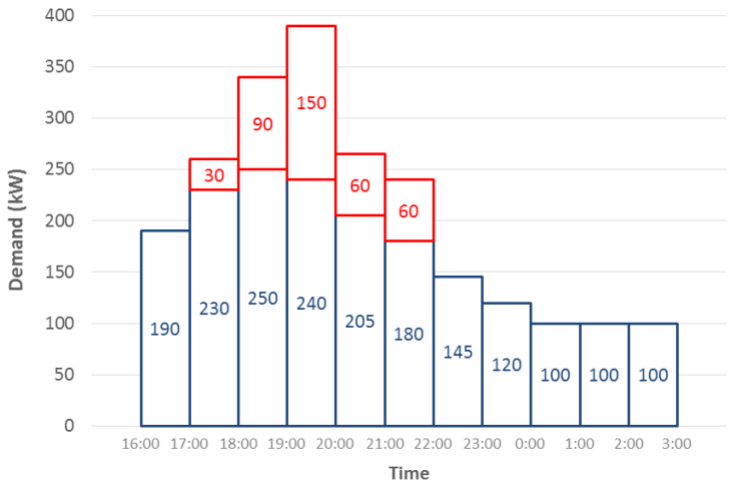
EVs in groups 1, 2 and 3 require 6, 12 and 18 kWh per vehicle, respectively, to charge their batteries to 100%. Respective charging times are 1, 2 and 3 hours when charging at rated power of the charger (6 kW). Multiplying the charger rating with the number of EVs gives the charging power for all EVs in group 1, 2 and 3 of 30 kW, 90 kW and 60 kW, respectively.

i) If all EV users plug in their vehicles to charge immediately upon arrival, and charge them at maximum capacity until their batteries are full, the resulting demand for each EV group as well as total demand are given in the table below.

Hour	Baseline	EV #1	EV #2	EV #3	Total EVs	Total demand
16:00	190					190
17:00	230	30			30	260
18:00	250		90		90	340
19:00	240		90	60	150	390
20:00	205			60	60	265
21:00	180			60	60	240
22:00	145					145
23:00	120					120
00:00	100					100
01:00	100					100
02:00	100					100

The new demand diagram including EVs is shown here:

Model Answers

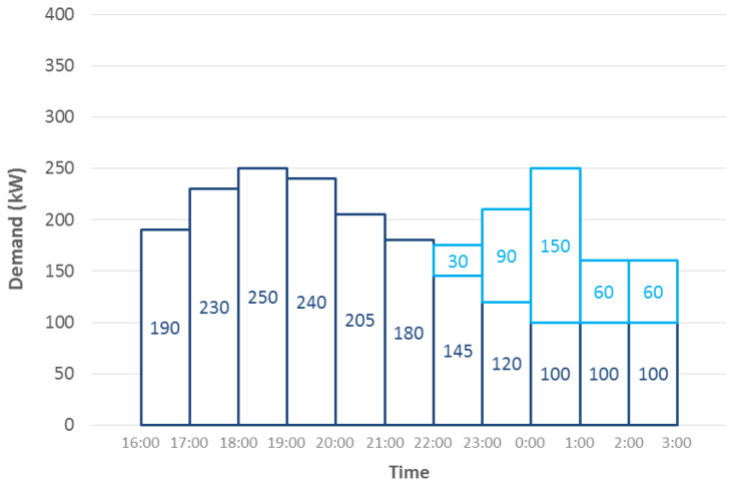


Peak demand is achieved in the hour between 19:00 and 20:00, at the level of **390 kW**.

b) By how many hours (at least) do all EVs need to delay their charging after arrival in order to avoid increasing the total peak demand of the feeder above the baseline peak demand of 250 kW? Assume that the same time delay for charging applies to all EVs.

[3]

To avoid increasing total peak demand, the hour with the maximum total EV charging demand of 150 kW (see figure above) needs to be shifted to 0:00 or later if the resulting peak should not exceed the 250 kW peak demand for the baseline profile. That means that all EVs need to delay their charging by at least **5 hours**. The resulting demand diagram after shifting EV charging by 5 hours is shown below, confirming that the peak demand is maintained at 250 kW.

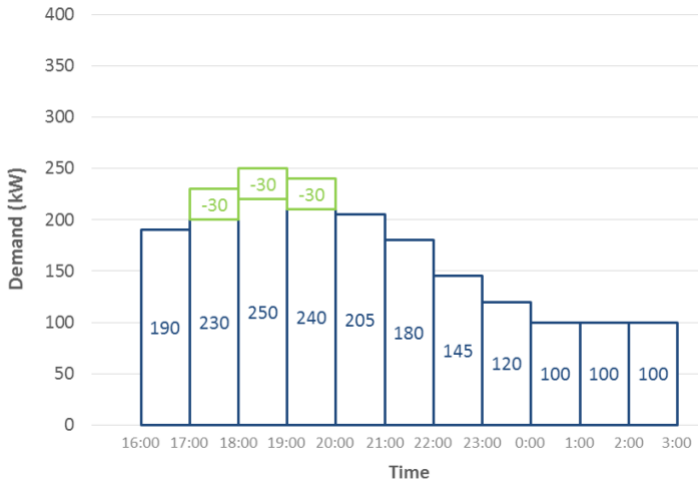


c) Assume that EV Group No 1 is willing to provide V2G services (i.e. discharge the batteries) between 17:00 and 20:00. By how much can the total peak demand of the feeder be reduced? For the purpose of this question assume that all EVs across all groups can be charged after 3:00 am without any impact on the total peak demand of the feeder.

[3]

MyAnswers

Discharging EVs in group #1 can inject 5*6 = 30 kW of power back into the grid. Doing this for 3 hours between 17:00 and 20:00 would require 18 kWh per EV, which is less than the assumed SoC on arrival (24 kWh), implying that this service is feasible. V2G reduces the demand in all three hours between 17:00 and 20:00 by 30 kW, so that the new peak demand occurs between 18:00 and 19:00 and is equal to 220 kW. Given that all EVs can be charged after 3:00am without any impact on the total peak demand, the new net peak demand is equal to **220 kW**. The resulting demand diagram is shown below.



Hydro Power

d) A hydropower station with a reservoir has an effective head of 120 m. The height of the surface of the headwater in the reservoir is 150 m. There are 20 generators each rated at 700 MW, 20 kV, 50 Hz with a rated speed of 100 rpm.

i) What is the height (in m) of the surface of tailwater?

[1]

$h = h_{hw} - h_{tw} \rightarrow h_{tw} = h_{hw} - h = 30$ m

ii) How many poles are there in each generator?

[1]

$$N\ (\text{in rpm}) = \frac{f\ (\text{in Hz}) \times 60}{p/_2}$$

$$\Rightarrow p = \frac{2 \times f\ (\text{in Hz}) \times 60}{N\ (\text{in rpm})} = 60$$

(problem solving – use of standard formulae and interpretation)

iii) What type of generator rotor construction is necessary to support the poles and how is it different from the rotor of thermal generators?

[2]

Model Answers

- Salient pole rotor construction is necessary to accommodate a high number of poles
- This is different from the cylindrical or round rotor construction used for thermal generators which operate a high speed (1500 or 3000 rpm) requiring only 2 or 4 poles

[1 mark for each point]

Some students strayed towards PMSG, converter configuration while some into Francis/Kaplan turbines etc. which are obviously irrelevant for this question

(bookwork - interpretation of material from lecture notes)

iv) How much discharge (in m^3/s) of water is necessary to produce the rated power of the hydropower station assuming 90% efficiency of the generator.

[1]

$P_{\text{in}} = \rho g Q h$
$P_{\text{out}} = N_g \times P_g$

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{N_g \times P_g}{\rho g Q h} \rightarrow Q = \frac{N_g \times P_g}{\eta \times \rho g h} = 13.21 \times 10^3 \text{ m}^3/\text{s}$$

Several students forgot to multiply P_g by N_g

(problem solving – use of standard formulae and deduction)

Offshore Transmission

e) Explain why HVDC is preferred for high voltage subsea cables beyond a certain distance (i.e. cable length).

[3]

- Cables have larger capacitance per unit length compared to that of an equivalent overhead line.
- Charging current drawn by the cable capacitance increases with cable length (due to larger total capacitance) and operating voltage ($|i_C| = \omega CV$). For long high-voltage cables, charging current could be prohibitively high and limiting the active current carrying capacity of the cable.
- Use of HVDC virtually eliminates the above problem of charging current and thereby, retaining the current (and hence active power) transfer capability of high-voltage long-distance cables.

[1 mark for each point]

Model Answers

Some students strayed into the general AC vs DC arguments without specific reference to sub-sea cables.

(bookwork – interpretation of material from lecture notes)

f) List two problems associated with the use of LCC HVDC for connecting remote offshore wind farms. [2]

- LCC HVDC requires a strong AC system and thereby cannot support the stat-up of the offshore wind farm
- LCC HVDC converter station requires a large footprint for reactive compensation and harmonic filtering which is not difficult to accommodate on a compact offshore platform
- Due to DC voltage polarity reversal, use of stronger and lighter polymeric cables which are suitable for subsea installation is not possible with LCC HVDC

[1 mark each for any two points]

(bookwork – interpretation of material from lecture notes)

Model Answers